Brian Kipkemoi

Graduate Electrical & Telecommunications Engineer — Wireless Networks · Communication Systems · Connected Infrastructure

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PROFILE

Moi University B.Eng. graduate in Electrical & Telecommunications Engineering with academic and practical exposure to communication systems, wireless networks, RF fundamentals, digital signal processing, data networks, transmission systems and embedded connectivity. Through university projects, laboratory work and a 12-week industrial attachment, I analyze communication problems, configure and test basic network systems, interpret signals and measurements, and document technical findings clearly. Open to graduate telecommunications, networking, RF and infrastructure roles.

EDUCATION

Moi University — School of Engineering, Eldoret, Kenya

B.Eng. Electrical & Telecommunications Engineering · 2021 – 2026 (expected)

Core coursework: Communication Systems · Digital Communications · Telecommunication Networks · Microwave Engineering · Optical Communication · Antennas & Propagation · Mobile Communication · Digital Signal Processing · Computer Networks · Embedded Systems · Circuit Theory · Control Systems · Electromagnetic Fields · Engineering Mathematics & Design.

TECHNICAL SKILLS

Communication Systems	Communication theory (strong) · analog & digital modulation · multiplexing · noise analysis · link budgets · error control concepts
RF & Wireless	Free-space path loss (strong) · propagation · antenna fundamentals · cellular architecture · spectrum concepts · LTE/5G (academic)
Networking	TCP/IP & IPv4 subnetting (strong) · routing & switching · VLANs · DHCP/DNS/NAT · Wireshark analysis · troubleshooting
Transmission	Fiber optics · optical power budgeting · multiplexing · microwave (academic) · link availability concepts
Signal Processing	Sampling (strong) · Fourier analysis · digital filtering · spectral analysis · MATLAB DSP
Embedded / IoT	Arduino & ESP32 (strong) · sensors · UART/I2C/SPI · MQTT · embedded C/C++ · AT commands

KEY ENGINEERING PROJECTS

• LoRa-Based Rural Environmental Monitoring & Long-Range Telemetry System — Final-Year Project (2026)
ESP32 sensor nodes + LoRa SX1278 (868 MHz) forwarding to a gateway and MQTT dashboard. Link budget ≈ 42 dB; field tests measured RSSI/SNR and packet delivery to 5 km (PDR 95.5% @ 2 km). Role: system design, embedded development, field testing.

• Enterprise Campus Network Design & Simulation — Academic Team Project (2025)
Designed and verified a 5-VLAN, 6-subnet campus network in Cisco Packet Tracer: inter-VLAN routing, OSPF, DHCP scopes, ACL concept, 42/42 connectivity test matrix. Role: IP addressing & VLAN lead.

• Digital Modulation Simulation & BER Analysis — Academic Project (2024)
MATLAB/Simulink simulation of BPSK, QPSK and 16-QAM over AWGN; BER-vs-Eb/N0 curves matched theory; quantified the ≈4 dB Eb/N0 price of 16-QAM's 4× spectral efficiency.

• Fiber Optic Link Power Budget Analysis — Academic Project (2024)
Proved a 12 km single-mode 1310 nm link closes with 12.4 dB margin (7.6 dB total loss); computed ≈42 km maximum reach for the selected optics.

• Campus Wi-Fi Coverage Survey & AP Placement Study — Field Project (2025)
42 reference points surveyed; identified two dead zones below –85 dBm and 3-way co-channel overlap; delivered an evidence-based AP placement and channel plan.

• GSM / SMS Remote Monitoring Prototype — Personal Project (2023)
ESP32 + SIM800L alert node with AT-command state machine, rate limiting and power conditioning; end-to-end SMS alert latency ≈ 14 s.

INDUSTRIAL ATTACHMENT

Telecommunications Engineering Intern — Network Operations / Transmission / Field Engineering

Telecom & network infrastructure company (name on request) · Kenya · May – Aug 2025 · 12 weeks, supervised

- Assisted with network monitoring, alarm observation and basic router/switch configuration under supervision.
- Participated in fiber and transmission troubleshooting; used ping, traceroute and CLI diagnostic tools.
- Observed base-station site procedures, preventive maintenance and escalation workflows for service-impacting faults.
- Assisted with IP addressing, connectivity tests, equipment inventory and fault documentation.

TOOLS

MATLAB / Simulink · Cisco Packet Tracer · Wireshark · GNS3 · Arduino IDE · PlatformIO · Proteus · LTspice · Python · Linux networking tools · Microsoft Excel · PuTTY

CURRENTLY DEVELOPING

CCNA-level routing & switching (IPv6) · LTE/5G architecture · GNU Radio & SDR · network automation with Python · LoRaWAN · cloud networking fundamentals.